

Allen X. Liu

Personal Information

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Website: <https://aliu42.github.io/>

Academic Appointments

New York University, Courant Institute

Assistant Professor of Computer Science, (Jan. 2026-present)

UC Berkeley

Miller Postdoctoral Fellow (Fall 2025)

Education

Massachusetts Institute of Technology

Ph.D in Computer Science, 2025

Advisor: Ankur Moitra

Massachusetts Institute of Technology

S.M. in Computer Science, 2022

Advisor: Ankur Moitra

Massachusetts Institute of Technology

B. Sc. in Mathematics, 2020

Awards and Honors

Quantum Information Processing (QIP) Best Student Paper (2024)

Citadel GQS PhD Fellowship (awarded 2024)

Hertz Fellowship (awarded 2021)

NSF Graduate Research Fellowship (awarded 2020)

International Mathematical Olympiad (IMO): Gold medalist (2014-2016), Perfect score (2016)

Working Experience

Research Intern, Microsoft Research, WA, Summer 2021, 2022 (hosted by Jerry Li)

Research Intern, Google Research, NY, Summer 2019 (hosted by Jon Schneider and Renato Paes Leme), Summer 2020 (hosted by Morteza Zadimoghaddam)

Quantitative Research Intern, D. E. Shaw & Co., NY, Summer 2018

Trading Intern, Jane Street Capital, NY, Summer 2017

Selected Publications

* denotes equal contribution, otherwise author order is generally alphabetical also denoting equal contribution

Subliminal Effects in Your Data: A General Mechanism via Log-Linearity

I. Aden-Ali*, N. Golowich*, A. Liu*, A. Shetty*, A. Moitra, N. Haghtalab
International Conference on Machine Learning (ICML 2026)

Sequences of Logits Reveal the Low Rank Structure of Language Models

N. Golowich*, A. Liu*, A. Shetty*

*International Conference on Learning Representations (ICLR 2026), **Oral Presentation***

Provably Extracting the Features from a General Superposition

A. Liu

Manuscript

Provably Learning from Modern Language Models via Low Logit Rank

N. Golowich, A. Liu, A. Shetty

Manuscript

Model Stealing for Any Low Rank Language Model

A. Liu, A. Moitra

ACM Symposium on Theory of Computing (STOC 2025)

High-Temperature Gibbs States are Unentangled and Efficiently Preparable

A. Bakshi, A. Liu, A. Moitra, E. Tang

IEEE Symposium on Foundations of Computer Science (FOCS 2024)

Structure Learning of Hamiltonians from Real-Time Evolution

A. Bakshi, A. Liu, A. Moitra, E. Tang

IEEE Symposium on Foundations of Computer Science (FOCS 2024)

Learning Quantum Hamiltonians at Any Temperature in Polynomial Time

A. Bakshi, A. Liu, A. Moitra, E. Tang

*Quantum Information Processing (QIP 2024), **Invited Plenary, Best Student Paper***

ACM Symposium on Theory of Computing (STOC 2024)

Invited to SIAM Journal on Computing Special Issue

An Optimal Tradeoff between Entanglement and Copy Complexity for Quantum State Tomography

S. Chen, J. Li, A. Liu

ACM Symposium on Theory of Computing (STOC 2024)

Constant Approximation for Individual Preference Stable Clustering

A. Aamand, J. Chen, A. Liu, S. Silwal, P. Sukprasert, A. Vakilian, F. Zhang

*Advances in Neural Information Processing Systems (NeurIPS 2023), **Spotlight***

When Does Adaptivity Help for Quantum State Learning?

S. Chen, B. Huang, J. Li, A. Liu, M. Sellke

Quantum Information Processing (QIP 2023)

IEEE Symposium on Foundations of Computer Science (FOCS 2023)

The Full Landscape of Robust Mean Testing: Sharp Separations between Oblivious and Adaptive Contamination

C. Canonne, S. Hopkins, J. Li, A. Liu, S. Narayanan

IEEE Symposium on Foundations of Computer Science (FOCS 2023)

Invited to SIAM Journal on Computing Special Issue

Matrix Completion in Almost-Verification Time

J. Kelner, J. Li, A. Liu, A. Sidford, K. Tian

IEEE Symposium on Foundations of Computer Science (FOCS 2023)

Semi-Random Sparse Recovery in Nearly-Linear Time

J. Kelner, J. Li, A. Liu, A. Sidford, K. Tian

Conference on Learning Theory (COLT 2023)

Tensor Decompositions Meet Control Theory: Learning General Mixtures of Linear Dynamical Systems

A. Bakshi, A. Liu, A. Moitra, M. Yau

International Conference on Machine Learning (ICML 2023)

A New Approach to Learning Linear Dynamical Systems

A. Bakshi, A. Liu, A. Moitra, M. Yau

ACM Symposium on Theory of Computing (STOC 2023)

Robust Voting Rules from Algorithmic Robust Statistics

A. Liu, A. Moitra

ACM-SIAM Symposium on Discrete Algorithms (SODA 2023)

Minimax Rates for Robust Community Detection

A. Liu, A. Moitra

IEEE Symposium on Foundations of Computer Science (FOCS 2022)

Tight Bounds for Quantum State Certification with Incoherent Measurements

S. Chen, B. Huang, J. Li, A. Liu

Quantum Information Processing (QIP 2023)

IEEE Symposium on Foundations of Computer Science (FOCS 2022)

Robust Model Selection and Nearly-Proper Learning for GMMs

J. Li, A. Liu, A. Moitra

Advances in Neural Information Processing Systems (NeurIPS 2022)

Clustering Mixtures with Almost Optimal Separation in Polynomial Time

J. Li, A. Liu

ACM Symposium on Theory of Computing (STOC 2022)

SIAM Journal on Computing Special Issue 2024

Learning GMMs with Nearly Optimal Robustness Guarantees

A. Liu, A. Moitra

Conference on Learning Theory (COLT 2022)

Settling the Robust Learnability of Mixtures of Gaussians

A. Liu, A. Moitra

ACM Symposium on Theory of Computing (STOC 2021)

Journal of the ACM 2023, 70(3), pp. 1-53

Variable Decomposition for Prophet Inequalities and Optimal Ordering

A. Liu, R. Paes Leme, M. Pal, J. Schneider, B. Sivan
ACM Conference on Economics and Computation (EC 2021)

Optimal Contextual Pricing and Extensions

A. Liu, R. Paes Leme, J. Schneider
Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2021)

Tensor Completion Made Practical

A. Liu, A. Moitra
Advances in Neural Information Processing Systems (NeurIPS 2020)

Better Algorithms for Estimating Non-Parametric Models in Crowd-Sourcing and Rank Aggregation

A. Liu, A. Moitra
Conference on Learning Theory (COLT 2020)

Fourier and Circulant Matrices are not Rigid

Z. Dvir, A. Liu
Computational Complexity Conference (CCC 2019)
Theory of Computing Special Issue 2020, 16(1), pp. 1-48

Efficiently Learning Mixtures of Mallows Models

A. Liu, A. Moitra
IEEE Symposium on Foundations of Computer Science (FOCS 2018)

Teaching

Teaching Assistant for 6.S896 (Algorithmic Statistics) at MIT, Fall 2023
Teaching Assistant at USA Math Olympiad Summer Program, Summer 2017, 2018

Service

Reviewer for SODA 2025, NeurIPS 2024, FOCS 2024, STOC 2024, ICALP 2024, FOCS 2023, STOC 2023, SODA 2023, FSTTCS 2023, FOCS 2022, SODA 2022, NeurIPS 2021, FOCS 2021,

Chief Problem Writer for Harvard MIT Mathematics Tournament (HMMT) 2017, 2018